Lume Centrix Framework

A Programmable Community Intelligence Layer provided as a platform template to Community Custodians

Requirements Specification

1. Identification

Project Name: Lume Centrix Framework

Product Class: Agentic Content-Aggregation Framework (ACAF)

Runtime: Hybrid (Web + Telegram Native)

2. Executive Technical Summary for Stakeholders

The Lume Centrix Framework is a multi-tenant, enterprise-grade SaaS platform designed to transition online communities from chaotic social media feeds into highly curated, AI-driven "Intelligence Hubs." It eliminates the need for manual community management and custom app development by utilizing a sophisticated, autonomous architecture broken into four core pillars:

1. The 7-Agent Intelligence Pipeline

   Unlike standard RSS readers, Lume Centrix employs a continuous loop of specialized AI agents. The Centrix Layer autonomously ingests, transcribes (via Whisper), and translates raw multi-modal media (video, audio, text). The Lume Layer then reformats this data into clean Markdown summaries while predicting sentiment and tracking the political leaning/objectivity of the source publishers.

2. Polymorphic Board Architecture

Hubs are not restricted to basic vertical newsfeeds. Community Custodians can configure up to 10 distinct boards utilizing modular Archetypes—seamlessly transitioning the UI from standard news streams to interactive event calendars, static entity registries, or transactional community marketplaces.

3. Dual-Runtime Delivery (Web & Telegram)

   To eliminate onboarding friction, the framework utilizes a hybrid delivery model. Casual users interact with a beautifully responsive, SEO-optimized web application (Tier 1). However, premium subscribers unlock the Tier 2 Runtime: a native Telegram Mini App. This provides the Hub with 100% reliable push-notification reach, bypassing algorithmic suppression from big tech platforms.

4. Decentralized Curation & Monetization

   The platform transforms passive readers into an active collective intelligence. Users dynamically train their own personalized "For You" algorithms, directly rate the credibility of media establishments, and submit content via a dedicated messaging inbox. Furthermore, Custodians can seamlessly monetize this high-trust environment via a native Ad Console and premium membership gating.

The SaaS Deployment Model & Collective Intelligence Loop As a white-label, multi-tenant SaaS solution, the framework operates on a single, massively scalable infrastructure. When a Community Custodian registers, they are instantly provisioned a secure, logical "Workspace" (referred to as a Hub) that maps directly to their own custom domain.

Ultimately, this deployment model elevates the Hub from a simple news aggregator into a Collective Intelligence Engine. By combining Agentic Ingestion with community peer-review, the platform creates a mission-led filter for niche audiences driven by a continuous, self-optimizing feedback loop:

- The Intelligence Loop: `Raw Internet Data → AI Agents (Ingest/Summarize) → Polymorphic Boards → Users (Consume/Rate) → AI Agents (Refine & Predict)`

This loop guarantees that the longer a Hub operates, the smarter, more objective, and more personalized its curation becomes.

3. High-Level Architecture: The "Four-Layer" Logic

3.1 Terminology

Monitored Sources: the Monitored Sources list is a curated list of content producers - e.g., media establishments like YouTube, Rumble and Vimeo channels, blogs, websites, social media pages, etc. - that are considered to be producers of content that is deemed to be relevant for the particular Hub that draws content from them. Can also be thought of as Publishers or (Content) Feeders.

Hubs: a Hub is an isolated, bespoke instance of the Lume Centrix framework deployed for a specific community custodian.

Boards: The primary "pages" configured for a Hub (up to 10 per Hub) that each present a certain type of content, including the Feed Filter and tools utilized by the board.

Board Type (Archetypes): The structural classification assigned to a board that strictly governs its UI layout (e.g., a vertical feed, a chronological calendar, or a static registry grid), the specific Card Types it supports, its available Action Bar tools, and its underlying data ingestion logic (external monitored sources vs. internal system data).

Embedded Boards: Sub-boards that exist entirely within the context of a parent Card (e.g., a comment stream), inaccessible from global navigation.

Cards: The standardized UI container for Monitored Sources Publications (MMPs), existing in Contracted or Expanded states.

Card Types: The platform template provides **four foundational Card Types** based on their data schema and presentation layout, viz., Narrative Cards, Chronological Cards, Entity Cards, and Transactional Cards.

Board Content Display: The primary viewing area (typically the center column) in the UI that presents the board's populated content. By default, it functions as an infinite-scroll feed of minimal, contracted **Tiles** (displaying only essential metadata such as a title, byline, publisher, category tag, timestamp, and a thumbnail or featured hero image). These tiles serve as interactive gateways that, upon user selection, expand inline into fully detailed **Expanded Cards** to reveal the AI-generated summaries and the Action Bar..

Taste Graph: A continuously updated vector representation of a user's or community's preferences, driven by their ratings and interactions, used to algorithmically rank and recommend cards).

Feed Filter: The left-hand navigation column that allows users to dynamically filter a board's content by specific Publishers or source tags.

3.2 Platform Business Model

To sell this as a "*template*", we define it by its four functional layers:

(note: while the framework uses "Boards" internally, these can be presented to users as "Channels" in the UI)

Layer	Component	Function
Management	The Admin Console (Control Plane)	The "Command Center" for the custodian. Used to configure boards, whitelist sources, "tune" agent personas, and manage paid placements. Includes the "Monitored Sources List" - a curated registry of approved publishers with "Publisher Profiling Management" and "User Taxonomy Control" (Casual vs. Registered vs. Admin); also manage monetized placements via the *Marketplace & Ad Console*.
Ingestion	Centrix Agentic Artefacts	7 specialized agents (Sourcing, Transcriber, Translator, Creative, Taste Predictor, Rating Profiler, Publisher Profiler) that monitor, transcribe, process, and mapping Taste Vectors ("pre-rank" multi-modal sources such as YT, RSS, Social), and the Publisher Profiler (responsible for the Establishment Scorecard).
Curation	The Lume Board Engine	A multi-tenant controller managing up to 10 configurable board repositories. It utilizes *Routing Rules* to push agent outputs into specific *Board Archetypes* (e.g., standard feeds, calendars, registries) and manages the *Establishment Scorecards*. Handles metadata, "Contracted" vs. "Expanded" card states, and Paid

		Placement injection. Manages "Publisher Records" and the "Action Bar" metadata (claps, ratings, comments, etc.).
Delivery (Tier 1)	Standard Runtime (PWA)	**Standard Browser Hub:** A browser-based Progressive Web App providing cross-platform access to "Boards." Focuses on SEO, guest browsing, and conversion hooks. Some agentic utility (Transcripts/Translations).
Delivery (Tier 2)	Elite Runtime (Telegram TMA)	**Premium Intelligence Layer:** Featuring 100% reliable push alerts, personalized "For You" feeds, and advanced API tools like the "Add to RAG" export.

3.3 SaaS Multi-Tenancy & Infrastructure Architecture

To ensure rapid market deployment and low friction for Community Custodians, the Lume Centrix Framework operates as a centralized, managed Software-as-a-Service (SaaS). All Hubs (instances) share the same underlying core application and agentic pipeline compute resources, necessitating strict protocols for data privacy, tenant isolation, and white-label branding.

3.3.1 Multi-Tenant Data Isolation (Logical Separation)

The framework must employ a robust logical isolation architecture to guarantee that data, user profiles, and agentic intelligence models do not "bleed" between different Hubs (e.g., ensuring *Volkwolk* data is entirely inaccessible to the *Carrier Pigeon* Hub).

- **Database Schema & Row-Level Security (RLS):** A strict single-database, multi-tenant architecture must be used. Every primary entity record in the relational database (Cards, Boards, Publishers, Users, Comments) must be tagged with a unique tenant_id (Hub ID). The database must enforce Row-Level Security (RLS) policies to ensure that queries can only read, write, or update records associated with the authenticated tenant_id.
- **Vector Database Namespace Isolation:** The agentic pipeline relies on vector databases for the Taste Predictor and Rating Profiler agents. To prevent the "Taste Graph" of one community from skewing the recommendations of

another, vector embeddings must be strictly partitioned. This must be achieved either by using dedicated Namespaces per tenant_id or by applying hard metadata filters on all vector similarity searches.

- **Agent Persona Isolation:** When processing content, the LLM-driven agents (Creative Agent, Taste Predictor) must dynamically load the specific "Seed Documents" and "Keyword Guardrails" assigned to the active tenant_id.

3.3.2 Custom Domain Mapping & White-Label Routing

To fulfill the "white-label" value proposition, the framework must allow Community Custodians to present their Hub under their own branded domain (e.g., www.volkwolk.com) without exposing the underlying Lume Centrix platform URLs.

- **Edge Routing Resolution:** The Delivery Layer must utilize edge network routing (e.g., Vercel Edge Network, Cloudflare for SaaS, or a custom reverse proxy). When a user visits a custom domain, the edge router must dynamically inspect the incoming hostname, look up the corresponding tenant_id in the management database, and serve the correctly branded PWA (Tier 1 Runtime) and board configurations for that specific Hub.
- **Automated SSL/TLS Provisioning:** The framework must automatically provision and renew SSL certificates for all custodian-mapped custom domains to ensure secure, HTTPS-encrypted access without requiring manual server configuration by the custodian.
- **Bot Identity Masking (Telegram Tier 2):** For the Elite Runtime, the tenant_id resolution dictates which Telegram Bot Token is used. While the backend logic is processed by the centralized Lume Centrix engine, the end-user interacts exclusively with a custom-named bot (e.g., @Volkwolk_Bot) managed via the custodian's own Telegram API credentials, maintaining the white-label illusion.

3.4 The Intelligence Layer (Agentic Pipeline)

The Lume Centrix framework operates on a continuous, 7-agent feedback loop. These specialized AI subsystems are responsible for the ingestion, formatting, and algorithmic scoring of all Hub content.

3.4.1 Ingestion & Processing Agents (The "Centrix" Layer)

This subset handles the initial extraction and cleaning of noisy, multi-modal internet data.

- **The Sourcing Agent**: Monitors the whitelisted "Monitored Media Registry" (e.g., X, YouTube, RSS feeds). It acts as the primary filter, discarding irrelevant noise and passing viable links into the pipeline.
- **The Transcriber Agent**: Triggered specifically for video (YouTube) or audio (Podcast) sources. It utilizes Speech-to-Text models (e.g., Whisper) to generate a complete text transcript, creating a readable base for downstream agents.
- **The Translator Agent**: A dynamic localization agent. If the ingested source is in a foreign language, this agent translates the raw text or transcript into the Hub's primary default language prior to summarization.

3.4.2 Formatting & Predictive Agents (The "Lume" Layer)

This subset generates the standardized presentation and pre-calculates algorithmic relevance.

- **The Creative Agent (The Editor)**: Takes the raw or translated text and distills it into the standardized Card schema. It generates the concise headlines, the 150-character contracted bylines, and the full Markdown-rendered summaries for the Expanded state.
- **The Taste Predictor Agent**: Performs pre-emptive sentiment and leaning analysis. Before a newly ingested card is viewed by any user, this agent evaluates the text and assigns baseline "Taste Vectors" (e.g., `#HighlyTechnical`, `#Neutral`, `#Optimistic`). This solves the "Cold Start" problem, allowing the system to recommend new content instantly.

3.4.3 Community & Feedback Agents (The Optimization Layer)

This subset actively learns from user behavior to drive the platform's personalization and credibility metrics.

- **The Rating Profiler (Collaborative Filtering Agent)**: Powers the personalized "For You" boards. It maps the interaction history (claps, saves, "Show Less Like This" actions) of User A against similar clusters of users. It continually adjusts the user's personal Taste Graph to surface highly relevant content.
- **The Publisher Profiler Agent**: Governs media credibility. This agent aggregates all explicit user ratings regarding a specific source over time (via the "Classify" tool). It dynamically calculates and updates the "Establishment Scorecard" for each publisher, tracking community consensus on their objectivity, political leaning, and historical accuracy.

*** **Developer Note:** This architecture ensures that you (the platform owner) only have to maintain and deploy one codebase and one instance of the AI Agent pipeline, drastically reducing overhead while still providing "bespoke" environments to your clients.

4. Standard Runtime (PWA) Intelligence Hub

This tier 1 option is a Progressive Web App (PWA) which provides the standard browser-based hub for Lume Centrix Framework. It functions as the accessible, cross-platform entry point for all users—especially casual/guest visitors, new users during onboarding, and anyone preferring a full-screen web experience without relying on Telegram.

It is a fully responsive, installable web application that users access by visiting the community's branded domain (e.g., carrierpigeonhaven.com). It behaves like a lightweight native app once installed, providing a polished, app-like experience directly in the browser.

4.1 Core Characteristics & Capabilities

- **Installable & App-Like Feel:**

  Users see an "Add to home screen" or "Install" prompt (native browser behavior when criteria are met: HTTPS, manifest, service worker). Once installed:

  - Standalone window mode (no browser UI chrome).

    - Custom app icon, name, theme colors, and splash screen matching the community's branding.
    - Launches from the device's home screen/app drawer like any other app.

- Responsive & Cross-Platform:

  Adapts seamlessly to desktop, tablet, and mobile viewports. Optimized layouts include:

    - Grid or card-based feed on larger screens.
    - Vertical scrolling list on phones with touch-friendly elements.

- Offline & Reliable Performance Powered by service workers:
    - Caches core assets (UI, manifests, common card thumbnails).
    - Allows viewing previously loaded boards/cards offline (with graceful degradation for fresh content).
    - Background sync for pending ratings/promotions when connectivity returns.

- SEO & Discoverability: Server-side rendering or static generation for initial pages ensures search engines can crawl boards, card summaries, and metadata. Public boards appear in Google searches, driving organic traffic to the community.

- Guest vs. Authenticated Access
    - Casual/Guest users: Browse public boards anonymously with read-only view (no ratings, promotions, or personalized ranking).
    - Registered users (via Google Sign-In): Full interactivity, including personalized "For You" board based on taste profile.

4.2 User Interface & Layout

The web-based user interface must prioritize clean content discovery, utilizing a structured layout focused on content consumption that balances global navigation with deep, contextual exploration, at the root level similar in style to the Brave News interface. The layout is defined by the following core UI components and behaviors:

4.2.1 Header

- Community logo and name
- Global Search & Discovery Engine

  The search bar positioned in the Header acts as the primary discovery tool for the Hub, extending far beyond simple title matching. It must support the following capabilities:

  - Full-Text Indexing: The search engine queries the complete depth of the card's data schema. This includes the original headline, the full Markdown-rendered summary generated by the Creative Agent, and any underlying text processed by the Transcriber or Translator Agents.
  - Faceted & Metadata Search: Users must be able to search using, or filter results by, the Hub's structured metadata. This includes searching for specific Publisher names, system-assigned classifications (e.g., `#International`), or agent-generated sentiment tags (e.g., `#HealthAlert`).
  - Contextual Scope (Global vs. Local): By default, the search operates globally across all boards the user has access to. However, the UI must provide a quick toggle or dropdown to restrict the search scope to the currently active board (e.g., searching only within the "Events" calendar) or strictly within the user's personal "Saved/Bookmarked" list.
  - Intelligent Result Presentation: Search results are presented in a dedicated, infinite-scroll overlay or a temporary "Search Results" board. Results are dynamically sorted by relevance (keyword density and date), with cards previously "Saved" or "Clapped" by the user receiving a slight visual highlight or ranking boost to reflect their personal Taste Graph.

- Manual Refresh Button (Soft-Sync Trigger)

  A dedicated UI control (typically a circular arrow icon) that allows the user to manually poll the server for new data without triggering a disruptive, full-page browser reload. Activating this button executes the following state updates:

    - New Content Injection: It queries the backend for any newly processed Monitored Media Publications that match the currently active board's routing rules or the user's Feed Filter settings. New contracted tiles are seamlessly prepended to the top of the infinite-scroll feed.
    - Metadata & Engagement Sync: It refreshes dynamic engagement metrics on currently visible cards, updating live counts for claps, comments, and Publisher Scorecard adjustments driven by recent community interactions.
    - Taste Graph Recalculation (Contextual): If the user is currently viewing the algorithmically sorted "For You" board, triggering the refresh forces the UI to fetch a newly sorted batch of cards, instantly reflecting any explicit or implicit feedback (e.g., "Show Less Like This" actions) the user just provided.
    - Scroll Reset: Activating the refresh automatically snaps the user's viewport back to the top of the active board to display the newly fetched content.

- Authentication, Registration & Membership Module

  The Header must include a dynamic user-state cluster that handles identity and Hub-specific monetization. Its presentation adapts based on the user's current access tier:

  - Unauthenticated State (Casual Users): Displays clear "Sign In" and "Join / Subscribe" actions.
    - *The Join Flow:* Clicking "Join" must trigger a unified onboarding modal or dedicated membership page. This interface dynamically presents the specific membership options configured by the Hub Administrator.
    - *Dynamic Pricing & Tiers:* The framework must support both Free Registration (Google Auth flow to become a "Registered User") and Paid Subscriptions (to become an "Elite User"). If the Admin has monetized the Hub, this UI will display the pricing blocks, billing

intervals, and feature unlocks (e.g., Telegram access, Push Alerts, "Add to RAG" tool) associated with the premium tiers.

  - Registered State (Free/Guest Users): Displays the user's profile avatar and a prominent "Upgrade" or "Get Elite" Call-to-Action (CTA). This allows users who joined for free to easily discover the Hub's premium offerings and hit a paywall/checkout flow.
  - Elite State (Premium/Subscribers): Replaces the upgrade CTA with an "Elite" badge or premium styling around the user's avatar, granting access to a dropdown menu for managing their subscription, RAG webhooks, and billing settings.

- How it Works page link: a prominent text link or "info" icon placed immediately adjacent to the "Join / Subscribe" and "Sign In" buttons in the top Header; this page serves as both an educational tool and a sales pitch for the Elite Tier (see section 4.2.7, How it Works).

4.2.2 The Top Horizontal Toolbar (Global Controls)

A persistent horizontal toolbar must be anchored to the top of the interface, providing the user with global navigation and board-specific control mechanisms.

- Board Navigation Menu: A dedicated icon/menu (e.g., a "Hub Menu" or "Grid" icon) that, when clicked, displays a drop-down or modal list of all available boards within the Hub (e.g., News, Publishers, Events). Clicking a board name immediately navigates the user to that board, making it the active view.
- Taxonomy & Sort Controls: For boards configured with specific taxonomies (e.g., categorized by genre, topic, or date), a "Sort" icon must be present in the toolbar. Clicking this icon reveals a menu of relevant sort parameters (e.g., Sort by Date, Sort by Relevance, Sort by Genre, etc.), allowing the user to dynamically reorder the active board's feed.

4.2.3 The Main Feed Layout (Two-Column Architecture)

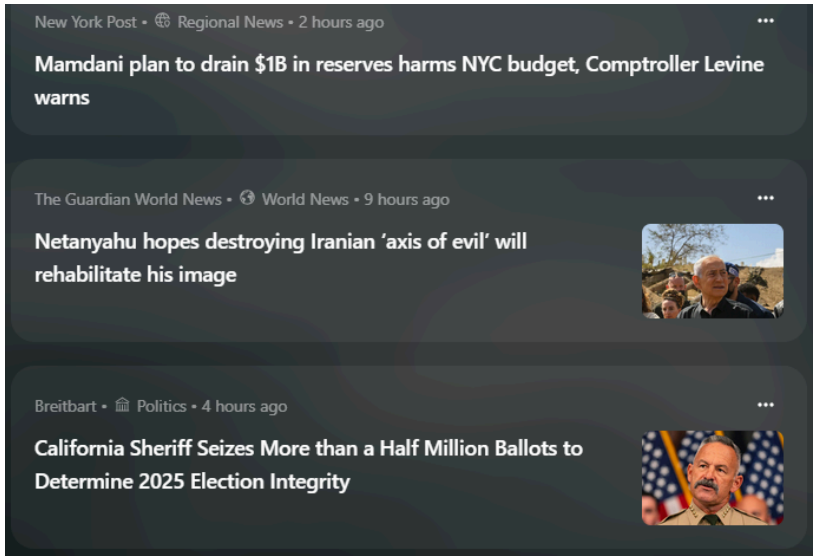
If we “divide” the presentation/display area horizontally into 4 equal parts, with two significant margins on either side of the display area, the first part of the 4

would be the left column. The 2nd and 3rd part must be combined to form the centre feed column. The remaining fourth part would be the right-most column. Similar to the Brave News presentation on a standard laptop layout:

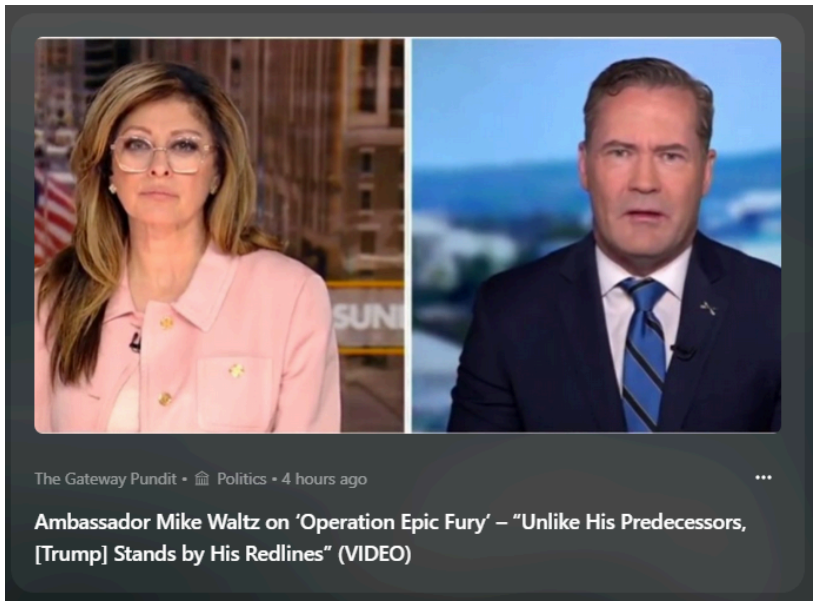
When viewing a standard content board (such as a news feed), the page must utilize a two-column layout:

- The Feed Filter (Left Column): A vertical sidebar located to the left of the main feed. This column dynamically lists all the Monitored Sources ("Publishers") that provide content for populating the active board.
    - Behavior: If a user clicks on a specific publisher in this list, the center feed column instantly filters to display *only* the cards sourced from that selected publisher (functioning similarly to the Brave News sidebar).
    - Board Selectors: - besides listing the names of the publishers, the Feed Filter column may also list the names of other boards, if the config settings for such “other board” has been set to “Show in Feed Filter”, which means the name of the particular board (with such setting set to active), will appear in the . styles & config Offers a navigation alternative to the Board Navigation Menu
- The Sourced Content Column (Center Double Column): The primary vertical scrollable area where the contracted "Cards" (MMPs) are stacked. The most recent or highest-ranked cards appear at the top, subject to the Sort Controls and Publisher Filters applied by the user.

- Infinite-scroll feed of content cards (default view for the selected board).
- 2 presentation styles for cards, viz., Standard Minimalist and Featured:



3 cards displayed in the Standard Minimilist presentation style



Single card displayed in Featured style

- The Featured presentation style is used to "feature" content that is popular, or rated highly relevant, or paid for. The presentation style for a particular card (Standard Minimilist or Features) is defined in the cards

metadata, with other elements (e.g., title, byline, source/publisher, publication date & time, etc.).

- Each card displays in Contracted state by default:
  - Eye-catching thumbnail (auto-selected by Creative Agent).
  - Optionally a Title (generated or refined by agents).
  - Byline or short summary (≈150 characters).
  - Feed object category (e.g., news, tech, entertainment, etc.)
  - Optionally source name + embedded profile link (clicking the publisher's name in the card opens a modal/side panel with a short summary of the configured profile, as per the profile elements the administrator selected in the publisher profile config, e.g., publisher's scorecard, political leaning, objectivity score, user ratings aggregate, etc. - link provided for navigating to the comprehensive profile - the "Publishers/Content Sources" board - if activated for the particular implementation of the template).
  - Metadata tags (e.g., #Rating #HealthAlert #Neutral); the visible outputs of the AI agents and the underlying data that drives the platform's intelligence, serving three critical functions:
    - Algorithmic Matching (The Content Vector): Tags generated by the Taste Predictor Agent (e.g., "Highly Technical," "Neutral") are combined with the source's reputation to create the "Content Vector". The Rating Profiler uses this vector to match the card against a user's personal Taste Graph for the "For You" board.
    - Contextual Visual Cues: They act as "Taste Graph indicators," providing users with subtle visual cues explaining *why* an article was recommended to them or what the AI determined its sentiment to be.
    - System Action Triggers: Certain tags act as routing rules. For instance, a #HealthAlert tag applied by the Sourcing Agent can trigger an immediate native push notification to Telegram subscribers.

        - Timestamp + priority/urgency indicator (if applicable); instead of timestamp it coud display “how long ago” it was published, e.g., “3 hours ago”.

- Card Expansion & Interaction
    - Tap/click a card → expands inline to Expanded state:
        - Full summary (Markdown-rendered, prepared by Creative Agent).
        - Optional embedded elements (e.g., transcript snippet from Transcriber Agent, translated text).
        - Original source link (external).

- Action Bar (horizontal icon bar at top or bottom of the card with smallish icons, each being a control that activates a particular tool). Tool availability is determined by the type of board (when the administrator defines a board, the admin would select which tools must be available to users of the boars; tools can be shared across board types. Tools are presented as small icons with the tool name displayed on hover. Tools include:
    - Open / Drill-down (Expand Card): Transitions the card from its default "Contracted Mode" to "Expanded Mode." This action expands the UI to reveal the complete, formatted output of the Creative Agent (e.g., full Markdown summaries), detailed metadata, the Publisher Scorecard, and extended interaction tools not visible in the contracted feed view.
    - Close / Go Back (Collapse Card): Reverses the "Open" action, collapsing the Expanded Mode back into the streamlined Contracted Mode. This restores the user's context within the vertical, scrollable board feed without losing their current position.
    - Promote / Clap (Positive Ranking Signal): An explicit positive feedback mechanism (implemented via a slider, multi-click button, or weighted toggle) that feeds directly into the Rating Profiler Agent. Activating this tool increases the algorithmic weight of the card's Content Vector,

instantly "training" the agent to prioritize and surface similar topics, tags, and sentiments higher up in the user's personal Taste Graph.

- Classify (Objectivity & Leaning Rating): Opens a structured rating interface allowing the user to evaluate the specific Monitored Sources Publication. This data feeds directly into the Publisher Profiler Agent, dynamically crowdsourcing the publisher's "Establishment Scorecard" by aggregating user assessments on metrics such as political leaning, objectivity, and potential misinformation.
- Comments (Embedded Discourse Board): Opens an "Embedded Board" (Sub-board) dedicated to user discussion specific to the active card. The icon displays a live count of current comments. When clicked, it allows Registered Users to submit text input, reply to peers, and engage in community-driven peer review of the source material.
- Tag / Save / Bookmark (Personal Curation): Enables the user to attach personal metadata to a card or add it to a persistent personal collection (e.g., "Read Later," "Favorites"). Applying this tool makes the specific card permanently retrievable via the Hub's global filter and search functionalities, regardless of the card's age or board ranking.
- Send / Share (Distribution): Invokes the device's native Web Share API or generates a copyable deep-link to distribute the card outside the Lume Centrix platform. Shared links are configured to route recipients back to the Hub's Tier 1 (PWA) runtime, acting as a top-of-funnel acquisition loop for new Casual Users.
- Listen (Text-to-Speech): Triggers an integrated Text-to-Speech (TTS) engine to read aloud the AI-generated summary of the card. This leverages the standardized output of the Creative Agent to provide accessibility and a hands-free "podcast-like" utility for users, which is particularly valuable in the mobile/PWA runtimes.
- Translate (Language Toggle): A dynamic UI toggle that instantly swaps the displayed text (titles and summaries) between the original source language and the Hub's (or user's) default language. This seamlessly

surfaces the pre-processed output generated by the framework's Translator Agent.

- Mute Publisher (Source Exclusion): A definitive, user-level hard-filter tool. Activating this instantly and permanently removes all past and future content originating from the specific Monitored Sources source from the user's personal feed. This operates independently of the community's Taste Graph, ensuring the source remains available to others while respecting the individual's boundaries.
- Report / Alert Admin (Moderation Flag): A security and moderation tool that flags the specific card (or a user comment) for manual review by a Hub Administrator. This is utilized for escalating Terms of Service violations, severe spam, explicit content, or critical ingestion errors made by the Sourcing Agents.
- Show Less Like This (Negative Feedback Signal): An explicit negative rating tool that feeds directly into the Rating Profiler Agent. Activating this tool instantly penalizes the algorithmic weight of the card's specific metadata tags and content vector within the user's personal Taste Graph. It serves to quickly "train" the agent to deprioritize, hide, or filter out similar topics, sentiments, or formats in the user's future feed generations.
- Add to RAG (Knowledge Base Export): *[Elite Users Only]* A premium data-portability tool that allows the user to push the active card's full content directly into their personal Retrieval-Augmented Generation (RAG) repository or external knowledge base (e.g., Notion, Obsidian, Custom GPT endpoint, or private vector database).
  - *Mechanism*: Activating this tool triggers a background webhook or API POST request configured in the Elite User's profile settings.
  - *Payload*: The system transmits a structured JSON payload containing the raw source text (or transcript), the Creative Agent's Markdown summary, the source URL, and all associated metadata tags (including Publisher Scorecard data and Taste Predictor

sentiment), providing clean, pre-processed data for the user's personal AI systems.

- Goto (Dynamic Contextual Navigation): A context-aware navigation tool that provides one or more routing options based on the specific board's configuration and the entity presented on the card. Because board relationships are configurable at the Hub level, this tool acts as a dynamic bridge:
  - *External Routing:* It may provide an "Outlink" to navigate the user away from the Hub to the original Monitored Sources Publication (e.g., viewing an article on the publisher's actual website).
  - *Internal Routing:* It may provide deep-links to other related boards within the Hub. For example, when viewing an "Associate Profile" card, the Goto tool may present a sub-menu allowing the user to jump directly to the Hub's "Events" board or "Opportunities" board, automatically filtered to show only items belonging to that specific associate.

- Hover-Summary / Peek (desktop): When a user hovers their cursor over a contracted card in the browser/PWA runtime, the UI must temporarily reveal a "Peek" overlay or expansion containing the following elements without requiring a full click:
  - The Full AI Summary: The contracted byline expands to display the complete, Markdown-rendered summary generated by the Creative Agent.
  - Publisher Snapshot: A lightweight tooltip or badge displaying the Publisher's "Establishment Scorecard" (showing their assessed political leaning and objectivity score).
  - The Quick-Action Bar: Temporary exposure of primary interaction tools (e.g., Promote/Clap, Save, Share), allowing the user to rate or bookmark the content directly from the feed without opening the fully expanded card view.

4.2.4 The Footer (Utility & Support Layer)

The Footer operates as a persistent utility and navigation layer anchored at the bottom of the Tier 1 (PWA/Desktop) runtime. While the Hub Administrator can customize specific URLs, the framework enforces a standardized layout comprising the following core components:

- Community Information (Hub-Specific)
  - About Us: A direct link to the Hub's foundational mission statement, the community rationale, or the expanded "How It Works" module.
  - Contact Us: A routing link configured by the Administrator (e.g., launching an email client, opening a contact form, or directing to a community Discord/Telegram channel).
- Resources & Documentation (The Qwiki System) The Footer serves as the entry point for user education and platform resources:
  - User Documentation (Qwiki): A link to the Hub's integrated "Qwiki" system. Qwiki is a lightweight, Markdown-based documentation engine hosted within the Hub. It acts as the central repository for all Hub-specific operational rules, community guidelines, and navigation tutorials.
  - Media & Updates: Configurable links to the Community Custodian's external or internal media channels, including a Vlog, Blog, and an interactive Product Tour for new user onboarding.
- Interactive Support: Ask.Snappy (AI Assistant)

  Rather than relying solely on static FAQs, the footer provides a trigger to launch Ask.Snappy, the framework's embedded conversational support agent.
  - UI Behavior: Clicking "Ask.Snappy" opens a persistent chat widget or floating modal.
  - Underlying Engine: Ask.Snappy is powered by a dedicated RAG (Retrieval-Augmented Generation) pipeline.

    - Knowledge Base: The RAG is dual-indexed. It is populated with the comprehensive *Lume Centrix Body of Knowledge (BOK)* to answer technical queries about platform navigation and features, as well as Hub-specific operational data to assist users with community-specific rules and workflows.

- Framework Attribution
    - "Powered by Lume Centrix": A mandatory, unobtrusive badge or text link acknowledging the underlying infrastructure. Clicking this link routes the user away from the active Hub and directs them to the overarching Lume Centrix public marketing site, serving as a top-of-funnel acquisition channel for new Community Custodians.

4.2.5 Contextual Shortcuts (Cross-Board Jumps)

The interface must support intelligent "Deep Linking" or "Jumps" between distinct boards, carrying contextual filters from the origin card to the destination board.

- Cross-Board Jump Logic: When viewing an expanded card or profile on one board, the UI must provide shortcut links to related boards.
- Example Application: If a user is on the "Publishers" board and opens the profile card for "XYZ Publishers" (viewing their Establishment Scorecard and ratings), a shortcut link labeled "View Publications" may be provided (depending on the config settings of the board). Clicking this link will then execute a jump to the primary "News" board, automatically applying a filter so the user only sees news cards published by XYZ Publishers.

4.2.6 Embedded Boards (Sub-Boards & The Tagging System)

To handle deep, hierarchical content without cluttering the top-level Hub navigation, the framework must support "Embedded Boards" (Sub-boards) that exist entirely within the context of a parent Card.

- Sub-Board Access: A prime example is the "Comments" section. Clicking the "Comments" icon on a parent card's Action Bar opens an Embedded Board consisting of user comments related specifically to that card. This sub-board is not accessible from the global Board Navigation Menu.

- The Tag Tool & Stream Following: Within an embedded board (like a comment stream), users must have access to a "Tag Tool." This allows the user to "Follow" or "Tag" that specific sub-board. (Note: If a user contributes a comment to the stream, the system automatically applies this Tag for them).
- Parent Board Filtering: Tagging a sub-board passes a metadata flag up to the parent card. The user can then use the Top Horizontal Toolbar on the main parent board (e.g., the News board) to filter the feed by "Followed/Tagged." This instantly narrows the main feed to show only the cards where the user is tracking the embedded sub-board discussions.

4.2.7 The "How It Works" / About Module

To bridge the educational gap regarding the Hub's agentic pipeline and to drive premium conversions, the framework must provide a dedicated, non-technical "How It Works" page.

- Link Placement & Accessibility Because this page serves as the primary conversion funnel for Casual Users, access to it must be highly visible and persistent across the UI:
  - The Header (Primary): A prominent text link or "info" icon placed immediately adjacent to the "Join / Subscribe" and "Sign In" buttons in the top Header.
  - The Feed Filter (Secondary): A persistent menu item located at the bottom of the left-hand navigation column (Feed Filter), sitting alongside user settings and legal/TOS links.
  - Empty States (Contextual): If a Casual User navigates to the "For You" board (which requires an account to function), the empty state must display a brief Call-to-Action linking directly to this page.
- Page Structure & Content Requirements The page must be presented as a visually engaging, scrolling landing page (or a seamless full-screen modal) rather than a dense wiki. It must contain the following modular sections:

- The Mission (Community Rationale): A customizable text block where the Community Custodian explains the specific purpose of their Hub (e.g., "Why we built this space for Tech Enthusiasts").
- The Concept (The "Engine" Explainer): A simplified, non-technical explanation of the Lume Centrix aggregation concept. It should emphasize time-saving and curation (e.g., *"Our AI agents monitor hundreds of trusted sources, transcribe videos, and summarize the noise so you only see what matters."*).
- Trust & Transparency (The Publisher Scorecard): A brief explanation of how the community keeps the feed objective. It must explain that human users (not just AI) rate the bias and credibility of the sources.
- The Value of Personalization (The Taste Graph): An explanation of why the user should register. It should outline how their interactions (claps, saves) train their personal algorithm to curate the "For You" board.
- Membership Tiers & Subscriptions (The Storefront): The culmination of the page. This section displays the dynamic pricing cards configured by the Administrator. It must clearly contrast the features of the Free/Registered tier (e.g., basic feed, standard summaries) against the Elite/Subscriber tier (e.g., Telegram Mini App access, Push Alerts, Text-to-Speech, RAG Export).

4.2.8 Personalized Elements & Feed Variations

While the Hub Administrator defines the global board topology, Registered and Elite Users experience personalized content routing. Depending on the Hub's configuration, users can engage with content across three distinct tiers of personalization:

- The Unified Community Board (Global):
  - Purpose: The baseline, shared community experience.
  - Behavior: This board displays the aggregated, unfiltered stream of all Monitored Media Publications approved by the Administrator for that

specific category (e.g., "All Tech News"). It relies on chronological or global popularity sorting rather than individual user preferences.

- The **User-Curated Board** (Explicit Personalization / "Following"):
  - **Purpose**: A strict, user-defined feed based on explicit choices.
  - **Behavior**: The user manually configures this board by "Whitelisting" or "Following" a specific subset of the community's available publishers or tags. For Elite Users, this board may also ingest custom, cross-community personal feeds (e.g., adding a personal RSS link that only the user sees). If a publisher is not explicitly approved by the user, their content will not appear here.
- The **"For You" Board** (Implicit / Algorithmic Personalization):
  - **Purpose**: A discovery-focused feed driven by the Sentrix AI pipeline.
  - **Behavior**: This board draws from the unified community pool but algorithmically re-ranks and filters the cards for the individual user. It utilizes the Taste Predictor and Rating Profiler to perform collaborative filtering based on the user's historical interaction data (claps, saves, time-on-card) and matches them against similar community clusters.

4.2.9 Additional Global Features

- Dark/light mode toggle (syncs with system preference).
- Notifications permission prompt (for web push of critical alerts, e.g., Health Alerts—falls back to in-app banners if denied).
- Onboarding tour for first-time visitors highlighting boards and actions.

*** **Implementation Note for the Developer**: The layout structure above represents the standard Tier 1 (PWA/Browser) runtime. The "Publisher Filter" and "Two-Column Layout" may be condensed into off-canvas sliding menus for mobile-sized browser viewports to preserve screen real estate.

5. The Telegram-Native "Mobile Runtime"

The Tier 2 Runtime utilizes the Telegram messenger platform as a native mobile UI host. Reserved for "Elite Users" (subscribers/authenticated members), this runtime eliminates the need for standalone iOS or Android app development while delivering 100% reliable push notifications and advanced agentic utilities.

The Telegram Runtime operates as a hybrid experience, utilizing both **Interactive Chat Messages** (for alerts and quick actions) and a **Telegram Mini App (TMA)** (for deep, visual browsing).

5.1 The Telegram UI Host Paradigm

Unlike traditional Telegram "Channels" (which are passive, one-way broadcasts), the Lume Centrix framework deploys an interactive Bot accompanied by a Telegram Mini App (TMA).

- **The TMA Canvas**: When the user taps a persistent "Open Hub" button inside the bot chat, a secure, web-based UI slides up over the messenger. This canvas mirrors the rich visual layout of the PWA but is heavily optimized for mobile touch interactions.
- **Identity Bridge (Instant Install)**: Users transition from the PWA to the Telegram Runtime via a secure "Magic Link" (utilizing the Telegram `start` parameter). This executes a handshake that links their Google SSO identity to their Telegram User ID, synchronizing their Taste Graph and saved content instantly.

5.2 Navigation & Board Access (TMA Layout)

Within the sliding TMA interface, the UI logic of the PWA is adapted for mobile constraints:

- **Board Navigation**: Instead of a top horizontal toolbar, board navigation (switching between News, Events, Opportunities) is handled via a sticky bottom navigation bar or a top-level swipeable tab menu.
- **The Publisher Filter & Sort**: The left-hand "Publisher Filter" from the desktop UI is converted into an off-canvas slide-out menu or a collapsible dropdown at the top of

the active feed, allowing users to quickly isolate content from specific Monitored Media entities. Sort controls (Date, Relevance) are housed in an adjacent icon.

5.3 Card Presentation & The Action Bar

Cards in the Telegram Runtime exist in two distinct states, leveraging Telegram's native capabilities:

- **In-Chat Push Cards (Contracted Mode)**: When the system pushes an urgent alert to the user, it arrives as a native Telegram chat bubble containing the contracted AI summary, thumbnail, and metadata tags.
- **Telegram Inline Keyboards (Quick Actions)**: Attached to the bottom of the in-chat push card is a Telegram Inline Keyboard (a row of native buttons). These buttons represent primary Action Bar tools (e.g., 👍 `Promote`, 👎 `Show Less`, 🔖 `Save`, and 📖 `Expand`). Tapping these buttons sends payload data directly back to the Rating Profiler Agent without opening the TMA.
- **TMA Expanded Mode**: When the user taps 📖 `Expand`, the TMA slides up, presenting the full Expanded Card. This view provides access to the complete Action Bar, including advanced agentic tools like **Listen (TTS)** and **Translate**, as well as the Publisher Scorecard.

5.4 Embedded Boards & Cross-Board Jumps

Deep linking and hierarchical navigation must function seamlessly within the Telegram environment.

- **Comments (Embedded Boards)**: Tapping the "Comments" tool within the TMA opens the contextual sub-board. Users can read and submit text replies directly within the TMA interface without being kicked out to a browser.
- **Cross-Board Jumps**: "Goto" actions function exactly as they do in the PWA. Clicking a jump link on an "Associate Profile" card will dynamically load the related "Opportunities" board inside the TMA, applying the appropriate associate filters instantly.

5.5 Push Intelligence & Agentic Alerts (Active Delivery)

The primary functional advantage of the Tier 2 Runtime is the "Active Delivery" mechanism driven by the Centrix Pipeline.

- **Critical Tag Routing**: The Admin Console can configure specific metadata tags (e.g., `#HealthAlert` or `#Urgent`) to bypass passive boards. When the Sourcing Agent applies one of these tags to a new card, the Bot actively pushes a native notification to the user's phone lock screen.
- **Personalized Threshold Alerts**: Because the Telegram Bot is synced with the user's Taste Graph, the Rating Profiler Agent can trigger silent or active push notifications when a newly ingested card achieves a high similarity score (e.g., >95% match) against the user's specific interests.

6. User Taxonomy & The Administrator Console

This section defines the hierarchical user access levels across the Lume Centrix Hub and the functional requirements for the Administrator Console, which acts as the centralized control plane for the Hub's agentic pipeline and board configurations.

6.1 User Taxonomy & Access Levels

To support a tiered engagement and monetization strategy, the framework must enforce the following four distinct user roles:

- **Casual User (Anonymous)**: An unauthenticated user accessing the Tier 1 Runtime (PWA/Browser). Casual users can only view "Public" boards, read contracted cards, and use the "Hover-Summary/Peek" function. They cannot utilize Action Bar tools that require identity (e.g., Promote, Classify, Comments, or Save) and do not have a personalized Taste Graph.
- **Registered User (Authenticated Guest)**: A user authenticated via a standard SSO provider (e.g., Google OAuth) accessing the Tier 1 Runtime. They have access to standard boards and a personalized feed. They can utilize all standard Action Bar tools (Promote, Classify, Tag, Comment). Their interactions actively train their personal Taste Graph and contribute to the community's Publisher Scorecards.

- **Elite User (Subscriber)**: A premium authenticated user with linked access to the Tier 2 Runtime (Telegram Mini App). Elite Users gain access to all restricted/premium boards, 100% reliable push alerts for critical tags (e.g., #HealthAlert), and full use of the advanced agentic tools (Listen/TTS, Translate).
    - **Personal Integrations (RAG Export)**: Elite Users are provided with an "Integrations" settings panel within their profile. This allows the user to define a target Webhook URL and an Authorization Header (API Key). Configuring this enables the **"Add to RAG"** Action Bar tool, allowing the user to seamlessly export cleaned JSON payloads (source text, AI summaries, and metadata) directly into their personal Retrieval-Augmented Generation (RAG) repositories or private knowledge bases.
- **Administrator (Community Custodian)**: A special class of user with elevated credentials granting access to the secure Administrator Console. Administrators do not just consume content; they dictate the ingestion, curation, and governance rules for the entire Hub.

6.2 The Administrator Console (Control Plane)

The Admin Console is a secure, web-based dashboard decoupled from the primary Hub UI. It provides the Administrator with no-code tools to orchestrate the 7-agent pipeline and manage the Hub's topology.

6.2.1 Source & Publisher Management

- **Monitored Sources Registry**: The Admin must be able to add, edit, or remove source endpoints (e.g., YouTube Channel URLs, RSS feeds, X handles, specific blogs).
- **Publisher Whitelisting:** The Admin must be able to approve or reject publishers, creating a curated "Monitored Sources List" that dictates which external entities the Sourcing Agents are allowed to pull from.
- **Source-to-Agent Routing**: The Admin must be able to explicitly assign ingestion agents to specific sources (e.g., toggling the Transcriber Agent ON for a newly added YouTube channel).

6.2.2 Board Typology & Presentation Architecture

The Lume Centrix Framework utilizes a polymorphic board architecture. The Admin must be able to define up to 10 distinct boards per Hub (e.g., News, Events, Opportunities). Boards are not restricted to standard vertical feeds of external media; rather, when an Administrator configures a new board, they must assign it a specific Board Archetype.

The selected Archetype strictly governs the board's visual presentation, the structural schema of its Cards, its available Action Bar tools, and its internal classification taxonomy.

6.2.2.1 Board Archetypes & Card Schemas

The framework supports the following core Board Archetypes, each utilizing a specialized Card structure:

- Standard Feed (News & Notifications): The default vertical scrollable list. Cards prioritize headlines, AI summaries, and Publisher Scorecards.
- Calendar / Chronological (events, projects, fairs): UI transitions to a date-based grid, timeline, or agenda view. The Card schema enforces mandatory start_date, end_date, and location metadata fields.
- Registry / Directory (Institutions, Associations, Profiles): A grid or list view optimized for static entities rather than time-sensitive news. Cards prioritize contact information, leadership profiles, and historical ratings.
- Marketplace (Opportunities, Offers, Help Requests): Optimized for transactional or action-oriented content. Cards feature explicit call-to-action buttons (e.g., "Apply," "Claim") and status flags (e.g., "Open," "Fulfilled").
- Interactive / Discourse (Chatgroups, Forums): Thread-based UI focusing on user-generated text inputs rather than external AI-summarized content.

- Personal / Gamified (My Tasks, "Helpende Hand"): User-specific private boards displaying personal assignments, saved notifications, or community reward metrics (tracking likes, views, and shares for gamified reputation).

6.2.2.2 Card UI Presentation Format

The underlying structural schema and UI presentation format of a Card, determined by its parent Board Archetype. The framework abstracts all content into four primary data structures:

- Narrative Card: Optimized for text-heavy content (e.g., News Articles, Notifications, Discourse). Prioritizes headlines, the Creative Agent's Markdown summaries, and publisher metadata.
- Chronological Card: Optimized for time-bound data (e.g., Events, Project Schedules, "Boerejolle"). Prioritizes `start_date`, `end_date`, `location` tags, and calendar-syncing tools.
- Entity Card: Optimized for static profiles and registries (e.g., Publisher Scorecards, Associate Profiles, Institutions). Prioritizes key-value attributes, avatars/logos, contact details, and historical ratings over narrative text.
- Transactional Card: Optimized for actionable listings (e.g., Marketplace Offers, Opportunities, Help Requests). Prioritizes explicit state flags (e.g., Open/Closed/Fulfilled), pricing or bounty metrics, and primary Call-to-Action (CTA) buttons.

6.2.2.3. Source Agnosticism and Content Routing

The framework handles data ingestion and board population differently based on the assigned Board Archetype and explicit routing configurations:

- External Monitored Media Boards: Populated autonomously by the Centrix Agentic Pipeline pulling from whitelisted publishers. For these boards, the Administrator must configure explicit Routing Rules to dictate where processed content is displayed:

    - *Publisher Origin Mapping:* Routing all content originating from a specific publisher (e.g., *Publisher A*) strictly to a designated board (e.g., the *News* board).
    - *Tag-Based Routing:* Dynamically routing content based on metadata tags assigned by the Sourcing Agent (e.g., automatically sending any card tagged with `#Events`, regardless of its publisher, directly to the *Community Calendar* board).
- Internal System Boards: Populated manually via Administrator input, user submissions, or internal API integrations (e.g., Registry boards or Gamified "My Tasks" boards). These bypass the external Sourcing Agents and external routing rules, but may still utilize the Rating Profiler and Translator agents for internal processing and presentation.

6.2.2.4. Intra-Board Classification (Taxonomies)

While the *Archetype* defines the board's overall structure, the *Classification System* defines how content is organized *within* that single board.

- Administrators can define a custom taxonomy for each board (e.g., configuring a News board with categories for International, Local, Economics, or an Events board with Online, In-Person, Hybrid).
- These classifications are presented to the user via the Top Horizontal Toolbar (as defined in Section 4.2.1) or the Feed Filter, allowing users to dynamically sort and isolate content within the active board.

6.2.2.5. Feature Limitations & Configuration

- The assigned Archetype hardcodes certain system limitations (e.g., an "Events Calendar" board cannot function without chronological metadata).
- However, within those constraints, the Administrator retains modular control to toggle specific features on or off (e.g., enabling or disabling the "Comments" Action Bar tool on an "Institutions Registry" board).

6.2.3 Agent Tuning & Governance

- **Creative Agent Prompting:** The Admin must be able to configure the summarization rules (e.g., setting maximum character limits for Contracted Mode, or defining the tone—such as "Academic" or "Casual"—for the Expanded AI Summaries).
- **Taste Predictor Seeding**: The Admin must be able to upload or paste a "Seed Document" (a text example of the community's ideal content). The Taste Predictor Agent uses this seed as the baseline to pre-rank and evaluate new, unrated content.
- **Translation Defaults**: The Admin must be able to define the Hub's primary default language and configure the Translator Agent to auto-translate specific foreign-language sources.

6.2.4 Analytics & Community Health

- **Publisher Scorecard Overview**: The Admin must be able to view the aggregated ratings of all Monitored Sources entities, seeing how the community classifies their political leaning and objectivity.
- **Taste Graph Clustering**: A visual or statistical dashboard allowing the Admin to see user sentiment trends (e.g., identifying if a large segment of the community is consistently hiding/downvoting a specific topic).
- **Moderation Queue**: A dedicated inbox where the Admin can review items flagged by users via the "Report / Alert Admin" tool, with options to delete the offending comment or ban the user.

6.2.5 Marketplace & Ad Campaign Console

To support platform monetization and sponsored placements, the Admin Console must include a dedicated module for managing paid "Ad Cards" within the Hub's organic feeds.

- **Campaign Manager:** The Admin must be able to create, schedule, and manage "Sponsored Cards." These cards mimic the UI of standard contracted/expanded cards but must be visually demarcated (e.g., tagged as "Sponsored") and bypass the standard ingestion pipeline.

- **Injection Ratio Engine:** The Admin must be able to define the display frequency of Sponsored Cards for each specific board. (e.g., configuring the "News" board to dynamically inject 1 Sponsored Card for every 8 organic cards as the user scrolls).
- **Taste Profile Targeting:** The Admin must be able to apply targeting rules to Sponsored Cards based on the Hub's aggregated metadata tags. This allows an ad to be shown only to users whose Taste Graph strongly aligns with specific topics (e.g., targeting a "Veterinary Services" ad strictly to users who frequently engage with #Health tags), ensuring high ad relevance without exposing or compromising individual user data.
- **Campaign Analytics**: A reporting dashboard for the Admin to track the performance of Sponsored Cards, specifically tracking impressions (views in the feed) and conversion clicks via the "Goto" Action Bar tool.

6.2.6 Outbound Syndication & Social Integration

To facilitate audience acquisition and external marketing, the Admin Console must support outbound webhook integrations to the Hub's external social media presences (e.g., X, Facebook Pages, Public WhatsApp Channels).

- **Syndication Triggers**: The Administrator must be able to configure rules that dictate which processed cards are automatically pushed to external channels. These triggers include:
  - *Tag-Based Auto-Posting:* Automatically syndicating cards that possess specific metadata tags (e.g., `#BreakingNews`).
  - *Threshold-Based Syndication:* Automatically syndicating cards that achieve a specific community engagement score (e.g., reaching 50 Claps).
  - *Manual Syndication:* A dedicated Administrator-only Action Bar tool allowing the manual push of an active card to connected social accounts.
- **Payload Formatting**: Outbound posts must automatically format to include the contracted headline, a truncated AI summary, and a direct referral link routing the external user back to the Hub's Tier 1 (PWA) runtime.

6.2.7 The Curation Inbox (User Submissions)

To support decentralized, community-driven content sourcing, the Admin Console must feature a dedicated "Curation Inbox."

- **Ingestion**: When Elite Users forward links, articles, or text directly to the Hub's connected WhatsApp or Telegram bots, the Sourcing Agent intercepts and structures the content, placing it in a "Pending" state within this inbox.
- **Admin Processing**: The Administrator can review the queue of user submissions. With a single click, the Admin can "Approve" a submission, triggering the rest of the agentic pipeline (Summarization, Taste Prediction) and routing the finalized card to the appropriate public board.
- **Attribution**: Approved cards generated via this inbox must include a metadata badge crediting the submitting user (e.g., *"Recommended by @UserHandle"*), incentivizing community participation.

6.3 Administrator Workflow Summary

The Administrator Console is designed to facilitate a logical, linear workflow for the Community Custodian. Setting up and managing a Lume Centrix Hub follows a standard five-step operational loop:

1. **Source Definition (The "Where")**: The Admin inputs raw source endpoints (YouTube channels, RSS feeds, X handles) into the Monitored Sources Registry and explicitly whitelists the approved publishers.
2. **Agent Assignment & Tuning (The "How")**: The Admin configures the intelligence pipeline by toggling specific agents for specific sources (e.g., turning on the Transcriber for video feeds), defining the default translation language, and uploading "Seed Documents" to calibrate the Taste Predictor.
3. **Topology & Routing (The "Where To")**: The Admin creates the Hub's visual structure (up to 10 boards) and maps the processed agent outputs to their respective boards. Additionally, the Admin configures Outbound Syndication rules, dictating which tags or highly-rated cards are automatically pushed out to the Hub's external social media channels.

4. Monetization Insertion (The "Revenue"): *(Optional)* The Admin accesses the Campaign Console to upload Sponsored Cards, define target Taste Profiles, and set the injection ratio for the organic feeds.
5. Monitoring & Governance (The "Iteration"): Once the Hub is live, the Admin continuously monitors the Analytics dashboard to review Publisher Scorecards, observe Tast*e Graph clustering, handle flagged items in the Moderation Queue, and approve or reject community-recommended links via the Curation Inbox—adjusting sources and agent prompts as the* community evolves.

7. Key System Workflows & User Journeys (Value Flows)

This section maps the chronological interactions between the system's Actors (Users, Administrators, and AI Agents) to illustrate the core value loops and operational sequences of the Lume Centrix Framework.

7.1 User Enrollment & Onboarding Flow

Actors: Casual Visitor → Registered User → Elite User(Subscriber)

Purpose: Establish identity, unlock interactive tools, and drive funnel conversion toward the premium Tier 2 runtime.

Key steps:

1. Discovery: The Casual User visits the Hub's custom domain (Tier 1 PWA) and browses public boards in a read-only state.
2. Registration: The user clicks "Sign In" (via Google Auth/SSO) to create a basic profile, becoming a Registered User. They gain access to the "For You" board and standard Action Bar tools.
3. Premium Upgrade (Optional): The user accesses the "Join/Subscribe" module to purchase a Hub membership.
4. Runtime Transition: Upon subscribing, the Elite User clicks a secure "Magic Deep Link" that instantly launches the Telegram Bot, synchronizing their web profile with their native Telegram identity and unlocking the Tier 2 Mini App.

7.2 Content Consumption & Interaction Flow (The Core Value Loop)

Actors: Registered User / Elite User

Purpose: Deliver personalized agentically intelligence and collect explicit feedback to train the predictive models.

Key steps:

1. **Discovery**: The user navigates to a specific board (e.g., *News* or *For You*).
2. **Scanning**: The user scrolls the infinite feed of minimalist Contracted Tiles.
3. **Expansion**: The user taps a tile (or hovers on desktop) to expand the card, revealing the Creative Agent's full Markdown summary and the Publisher Scorecard.
4. **Interaction**: The user utilizes the Action Bar to engage (e.g., applying the *Promote/Clap*, *Classify*, or *Show Less Like This* tools).
5. **Algorithmic Update**: The interaction payload is instantly routed to the Rating Profiler Agent, dynamically refining the user's personal Taste Graph and adjusting the community's baseline metrics for that specific publisher.

7.3 User-Submitted Content Flow (Collective Intelligence)

Actors: Elite User → Sourcing Agent → Administrator

Purpose: Enable premium community members to act as decentralized human ingestion sources.

Key steps:

1. **Submission**: An Elite User forwards an external article, link, or media file directly to the Hub's Telegram Bot chat interface.
2. **Agentic Validation**: The Sourcing Agent intercepts the link, filtering for malware, duplicates, or dead endpoints.
3. **Pipeline Processing**: If valid, the link is pushed through the standard Centrix pipeline (Transcriber, Translator, Creative Agent) to generate a standardized Card.

4. Publishing / Escalation: Depending on Admin routing rules, the card is either auto-published to a designated "Community Submissions" board, or placed in the Admin Moderation Queue for manual approval.
5. User Recognition

7.4 Profile & Integration Management Flow

Actors: Registered User / Elite User

Purpose: Empower users to customize their feed rules and configure external data portability.

Key steps:

1. Access: The user opens their Profile Dashboard via the Feed Filter menu.
2. Identity & Taxonomy: The user edits their biography, display name, and explicit topic interests.
3. Hard Filtering: The user manages their "Muted Publishers" list, explicitly blocking specific monitored sources from their feed.
4. RAG Configuration (Elite Only): The Elite User accesses the "Integrations" tab, inputting their target Webhook URL and API Authorization Header, permanently enabling the "Add to RAG" Action Bar tool.

7.5 Sponsored Placement Flow (Monetization)

Actors: External Advertiser → Administrator → Registered User

Purpose: Execute and manage natively integrated, non-intrusive monetization within the Hub feeds.

Key steps:

1. Advertiser accesses Advertiser Console (separate portal or admin-managed).
2. Admin reviews/approves via Marketplace & Ad Console.

3. Campaign Creation: The Administrator utilizes the Ad Console to upload a new campaign, defining the Sponsored Card's content, the target Taste Profile metadata (e.g., target users who follow `#Tech`), and the specific target boards.
4. Injection: The Injection Ratio Engine activates, seamlessly blending the Sponsored Card into the target boards at the defined frequency (e.g., 1 ad per 10 organic cards).
5. Conversion: A user matching the target Taste Graph views the feed, clicks the Sponsored Card's primary CTA ("Goto" tool), and routes to the external advertiser's landing page.
6. Performance tracked in Analytics.

7.6 Publisher Credibility Verification Flow

Actors: User (Any Tier) → Publisher Profiler Agent

Purpose: Enforce transparency and crowdsource the mitigation of media bias and misinformation.

Key steps:

1. Inspection: A user views a newly published card and clicks the Publisher Badge in the metadata line.
2. Review: A lightweight modal opens displaying the "Establishment Scorecard," revealing the community's aggregated assessment of the publisher's political leaning and historical objectivity.
3. Classification: A Registered User utilizes the "Classify" Action Bar tool to log their own rating of the source's bias on this specific article.
4. Recalculation: The Publisher Profiler Agent ingests the new rating, recalculating the publisher's overall credibility score in real-time.

7.7 Critical Alert & Push Notification Flow

Actors: Sourcing Agent → Telegram Bot → Elite User

Purpose: Deliver time-critical, mission-aligned intelligence directly to the user's device lock screen.

Key steps:

1. **Detection**: The Sourcing Agent ingests an article and flags it with an Admin-defined critical tag (e.g., `#HealthAlert` or `#Urgent`).
2. **Bypass**: The system routing rules recognize the tag, bypassing passive board placement.
3. **Active Push**: The Telegram Bot instantly pushes a native chat message to all Elite Users. The message contains the contracted summary, thumbnail, and inline keyboard buttons (Promote, Save, Expand) for immediate interaction without opening the full Mini App.

7.8 The Autonomous Ingestion Flow (Agent Hand-Off)

Actors: The 7-Agent Pipeline

Purpose: The fully autonomous, backend operational loop that translates raw internet data into intelligent UI Cards without human intervention.

Key steps:

1. **Detection**: A new YouTube URL is published by a whitelisted entity. The **Sourcing Agent** detects and extracts the media.
2. **Transcription**: The Sourcing Agent passes the audio track to the **Transcriber Agent** (Whisper), generating a raw text transcript.
3. **Localization**: The **Translator Agent** detects the transcript's language. If it differs from the Hub's default, it translates the entire text block.
4. **Summarization**: The **Creative Agent** ingests the text, writing the headline, crafting the 150-character byline, and generating the Markdown-formatted expanded summary.
5. **Pre-Scoring**: The **Taste Predictor Agent** scans the summary, assigning initial sentiment and topic tags (Taste Vectors) to solve the "cold start" problem.

6. Routing: The system evaluates the assigned tags and publisher origin, mapping the finalized Card to the appropriate Board Archetype based on Admin Routing Rules.
7. Personalization: Upon board load, the Rating Profiler Agent dynamically reorders the board's presentation layer for each specific user based on their historical interactions.

8. Non-Functional Requirements & Security

This section defines the system attributes, performance metrics, and security protocols required to ensure the Lume Centrix Framework operates as a scalable, reliable, and secure multi-tenant SaaS platform.

8.1 Performance & Latency

To maintain the illusion of a seamless "live" intelligence hub, the framework must adhere to strict processing and delivery baselines:

- UI Responsiveness: The Delivery runtimes (both Tier 1 PWA and Tier 2 Telegram Mini App) must achieve an initial Time-to-Interactive (TTI) of under 2.0 seconds on standard mobile broadband (4G/LTE).
- Search & Filter Latency: Global search queries, faceted metadata filtering, and "Refresh" actions must return populated results in under 500 milliseconds to ensure the UI feels instantaneous.
- Agentic Processing Timelines: * *Text-Based Content:* Standard text articles or RSS pulls must be ingested, summarized (Creative Agent), scored (Taste Predictor), and published to the active boards within 2 minutes of detection.
    - *Audio/Visual Content:* Video or podcast sources requiring the Transcriber Agent (e.g., Whisper) must be processed and published within 10 minutes of detection, depending on the source media's length.
- Push Notification Delivery: Critical tag alerts (e.g., `#HealthAlert`) routed via the Telegram bot must be delivered to Elite Users within 30 seconds of the card passing the ingestion pipeline.

8.2 Scalability & Concurrency

As a multi-tenant environment hosting numerous autonomous Hubs, the underlying architecture must scale dynamically without suffering from "noisy neighbor" degradation.

- Hub Concurrency: The framework database and agentic pipeline must independently partition and scale to support thousands of simultaneous Hubs without cross-tenant performance drops.
- User Concurrency: The Delivery runtimes must support a baseline of 10,000 concurrent active users per Hub without noticeable degradation in feed scrolling or Action Bar interactions.
- Agent Queueing & API Rate Limiting: The Sentrix Pipeline must implement an intelligent queuing mechanism. If upstream LLM providers (e.g., OpenAI, Anthropic) enforce rate limits, the pipeline must pool and throttle ingestion requests automatically rather than dropping data.

8.3 Availability & Resilience

- Uptime Target: The Tier 1 and Tier 2 Delivery runtimes, alongside the core Board database, must maintain a 99.9% high-availability uptime SLA.
- Graceful Degradation (Agent Failure): The framework must decouple the Delivery layer from the Intelligence layer. If the Agentic Pipeline goes offline (e.g., a third-party LLM outage), the user-facing Hubs must remain fully operational. Users must still be able to browse, read, and search previously processed cards, while new ingestions are safely queued in the background until the pipeline is restored.
- Disaster Recovery: Automated daily backups of all Hub configurations, Publisher Scorecards, and User Taste Graphs must be maintained, with a maximum Recovery Point Objective (RPO) of 24 hours.

8.4 Security, Privacy & Data Isolation

Because the framework processes personal interaction data and proprietary community sentiment, strict data governance is mandatory.

- Tenant & Vector Isolation: Hubs operate in strictly siloed logical workspaces. Under no circumstances can the Taste Graph vectors, Publisher Scorecards, or user data from *Hub A* bleed into or influence the AI algorithms of *Hub B*.
- Authentication Security: User authentication is handled strictly via secure OAuth 2.0/SSO providers (e.g., Google Auth) or native Telegram identity tokens. The framework will not store native user passwords.
- RAG Webhook Encryption: For Elite Users utilizing the "Add to RAG" tool, all external Webhook URLs and Authorization Headers (API Keys) must be stored using AES-256 encryption at rest.
- Data Ownership & Portability (No Lock-In): The Hub Administrator retains full ownership of their community's generated intelligence. The Admin Console must provide a one-click "Data Export" tool allowing the Custodian to download their community's Publisher Scorecards, structured Board schemas, and aggregated Taste Graph metrics in a standardized format (JSON/CSV) at any time.

9. Entity Relationship Diagram (Derived from Semantic Model)

9.1 Multi-Tenancy & Infrastructure

- **`HUB`**: The core tenant entity. Stores the community name, custom domain, default language, visual branding, and Admin configuration settings.

9.2 Identity & Access Management (IAM)

- **`USER`**: The global identity of a person across the entire Lume Centrix platform. Stores Google Auth ID, Telegram ID, global email, and base profile data.
- **`HUB_MEMBERSHIP`**: A linking table between `USER` and `HUB`. Stores the user's specific role in that specific Hub (e.g., Administrator, Elite, Registered), their billing status, and local profile bio.
- **`USER_INTEGRATION`**: Stores the Elite user's webhook configurations (e.g., the target URL and encrypted API key for the "Add to RAG" tool).

9.3 Source Management & Credibility (The "Where")

- **`PUBLISHER`**: The external media entity (e.g., "The New York Times", "Marques Brownlee").
- **`MONITORED_SOURCE`**: The specific endpoints the Sourcing Agent watches. Links to a `PUBLISHER` and a `HUB`. (e.g., Source Type: "YouTube Channel", URL: "youtube.com/c/mkbhd"). Includes agent toggles (e.g., Transcriber = ON).
- **`ESTABLISHMENT_SCORECARD`**: Stores the aggregated, dynamically calculated credibility metrics for a `PUBLISHER` within a specific `HUB` (e.g., Objectivity Score, Political Leaning index).

9.4 Content & Agent Outputs (The "What")

- **`RAW_INGESTION`**: The raw data pulled by the agents before it becomes a card. Stores original URLs, Whisper transcripts, and raw translated text.
- **`CARD`**: The central content entity (replacing your idea of POST/ENTRY). Stores the finalized Markdown summary, the contracted headline, thumbnail URL, and Card Archetype (Narrative, Chronological, Entity, Transactional).
- **`CARD_METADATA`**: A child table of `CARD` storing archetype-specific data (e.g., `start_date` and `location` for Chronological cards, or `price` for Transactional cards).
- **`TAG`**: The system-wide list of Taste Vectors and topics (e.g., `#HealthAlert`, `#Neutral`, `#HighlyTechnical`).
- **`CARD_TAG`**: A many-to-many linking table associating multiple `TAGs` to a specific `CARD`.

9.5 Topology & Routing (The "Where To")

- **`BOARD`**: A specific page within a Hub. Stores the Board Name, UI Archetype (Calendar, Feed, Registry), and active Action Bar toggles (e.g., "Comments Enabled = True").
- **`ROUTING_RULE`**: The Admin-defined logic that tells the system how to populate a `BOARD`. Evaluates `TAGs` or `PUBLISHER_IDs` to auto-route cards to the correct board.
- **`BOARD_CARD_MAPPING`**: While standard feeds might be populated dynamically via routing rules, this table explicitly tracks cards pinned, manually moved, or submitted to specific boards.

9.6 User Engagement & The Taste Graph

- **`TASTE_PROFILE`**: A mathematical vector representation of a user's preferences within a specific Hub, updated dynamically by the Rating Profiler agent.
- **`INTERACTION`**: Logs every user action on a `CARD`. Stores Interaction Type (Clap, Save, Read, Show Less), Timestamp, and the associated `USER_ID` and `CARD_ID`.
- **`CREDIBILITY_RATING`**: Explicit user votes submitted via the "Classify" tool. Tracks how a specific user rated a specific card's bias/objectivity (which the agents use to update the `ESTABLISHMENT_SCORECARD`).
- **`COMMENT`**: User discourse attached to a specific `CARD`.
- **`USER_SUBMISSION`**: Content submitted via Telegram/WhatsApp by Elite users, sitting in the Admin's Curation Inbox awaiting approval.

9.7 Monetization & Outbound Routing

- **`AD_CAMPAIGN`**: Admin-created configurations for Sponsored placements. Stores targeting rules (Target Tags), injection ratios, and active dates.
- **`SPONSORED_CARD`**: The actual advertisement payload (inheriting from the `CARD` structure but marked as sponsored).
- **`OUTBOUND_WEBHOOK`**: Admin rules for the Social Syndication feature. Stores external API keys (e.g., X, WhatsApp Channel) and the trigger conditions (e.g., Tag = `#Breaki`

Appendix A: Tier 2 Runtime (Telegram Bot & Mini App Integration)

A.1 Overview The Lume Centrix Framework utilizes a dual-runtime architecture. While the Tier 1 Runtime (PWA/Browser) serves as the top-of-funnel acquisition and standard consumption layer, the Tier 2 Runtime leverages the Telegram ecosystem to provide Elite Users (Subscribers) with a frictionless, high-retention mobile experience. This Tier 2

environment is strictly composed of two interconnected components: the Conversational Bot and the native Telegram Mini App (TMA).

A.2 The Bot Component (Push & Ingestion) The Telegram Bot acts as the asynchronous communication bridge between the Lume Centrix backend and the Elite User.

- **Authentication Handshake:** Registration is initiated via a securely signed "Magic Deep Link" generated in the Tier 1 PWA. Clicking this link launches the Telegram app and passes a unique authentication token (`/start=auth_[token]`), permanently mapping the user's web identity to their native Telegram User ID.
- **Active Push Delivery:** Bypassing algorithmic social media suppression, the bot delivers native chat messages for critical content. Governed by the Admin's Routing Rules, any card tagged with high-priority metadata (e.g., `#BreakingNews` or `#HealthAlert`) is immediately pushed to the user's Telegram chat thread.
- **Inline Interaction:** Push alerts delivered into the chat thread utilize Telegram's Inline Keyboard. This allows users to trigger lightweight Action Bar commands (e.g., "Expand Summary" or "Save to Profile") directly within the chat bubble without launching the full visual interface.
- **Decentralized Curation (Ingestion):** Elite Users can utilize the bot as a direct ingestion tool. By forwarding an external URL, article, or text payload to the bot, the submission is automatically routed to the Hub's Curation Inbox for Admin approval, effectively crowdsourcing content discovery.

A.3 The Telegram Mini App Component (The Visual Runtime) The Telegram Mini App (TMA) provides the full graphical user interface of the Hub, operating as a slide-up web-view securely sandboxed within the native Telegram mobile application.

- **Launch Mechanism:** The TMA is launched via a persistent, visually distinct "Menu Button" (e.g., "Open Hub" or "View Boards") anchored at the bottom of the Bot's chat interface.
- **Feature Parity:** The TMA perfectly mirrors the capabilities of the Tier 1 PWA. It renders the full Polymorphic Board Architecture, allowing the user to seamlessly scroll the contracted feed, view Entity Registries, and utilize the complete Action Bar (Promote, Classify, Comment).

- **Session Management & State Sync:** Because authentication is managed natively by Telegram's platform tokens upon launch, Elite Users never need to manually log in to the Mini App. Furthermore, all interactions (Claps, Saves, Profile updates) made within the TMA instantly sync with the central database, dynamically updating the user's central Taste Graph in real-time.

Key **Differences from Telegram Runtime (Elite Tier)**

Aspect	PWA (Standard Browser Hub)	Telegram Mini App (Elite Runtime)
Access	Any browser (URL or installed icon)	Inside Telegram (bot/chat list)
Push Notifications	Web push (requires permission)	Native Telegram push (100% reliable)
Offline	Partial (cached content)	Stronger caching + Telegram offline queues
Onboarding Friction	Google Auth → optional Telegram link	Instant after linking
Primary Strength	SEO, guest discovery, desktop experience	Mobile-first, alerts, zero-download install
Interaction Style	Full-page web UI, grid/list views	Chat-embedded, inline keyboards, slide-ups

In essence, the PWA is the open door to the Lume Centrix ecosystem—optimized for broad reach, easy onboarding, and a rich, visual browsing experience—while seamlessly funneling engaged users toward the Telegram premium layer for deeper, always-on intelligence.

Appendix B: Agentic System Integration

The platform's agents perform three main roles.

1 Source Monitoring Agents

Watch sources like:

- Facebook
- Instagram
- X
- Wikipedia
- YouTube
- Reddit
- Quora

Agents detect new items.

2 Card Generation Agents

Agents:

1. read content
2. summarize
3. generate card title
4. generate thumbnail
5. tag topic
6. assign board
7. determine ranking score

Card example:

Card ID

Title

Summary

Thumbnail

Board

Source URL

Publish Time

Expiry Time

Ranking Score

3 Ranking / Recommendation Agents

Use:

- user ratings
- similarity between users
- topic affinity

Example algorithm:

User vector = topics + ratings

Content vector = tags + source reputation

Recommendation score =

cosine_similarity(user,content)

8. Paid Placement Cards

Advertisers submit campaigns through the admin console.

Ad card example:

Sponsored

AI Bootcamp for Developers

Become job-ready in 12 weeks.

[Learn More]

Rules:

Board: Opportunities

Duration: 7 days

Frequency: every 10 cards

Geo-target: South Africa

The ranking engine injects them into feeds.

9. Subscriber Profiles

Profile contains:

User ID

Display name

Message / bio

Topic interests

Source preferences

Ratings history

Reputation score

The **rating profiler agent** builds:

User preference vectors

Cluster of similar users

Trust score

Influence score

This enables **collaborative filtering**.

10. Browser Version vs Telegram Version

Browser version behaves more like a **visual dashboard**.

Boards

News | Opportunities | Events | Notifications

Card grid layout

Cards expand inline.

Telegram version is **message-based navigation**.

11. Why Telegram Is Brilliant for This

Telegram gives you:

- push notifications
- chat UI
- bot automation
- identity
- global distribution
- zero mobile app maintenance

Essentially Telegram becomes your **mobile runtime environment**.

12. A Very Powerful Extension (Future)

Your boards could also become **public intelligence channels**.

Example:

African Tech Radar

SME Opportunities

AI Research Feed

Startup Funding Alerts

Communities subscribe to them globally.

APPENDIX C: Public Product Marketing Website

The public-facing Lume Centrix website serves strictly as the marketing and conversion portal for the framework itself. Its primary target audience is prospective **Community Custodians**. It is distinctly separate from the actual community Hubs (the various PWAs and Telegram Mini Apps created using the framework) and the Admin Console. Its sole purpose is to act as a narrative funnel: to educate visitors on the value of the agentic framework, demonstrate its capabilities, and convert them into platform administrators who sign up to deploy their own bespoke Lume Centrix Hubs.

When this public-facing website is created, the following guidelines and instructions that is provided in the “Web Site/App Design Guidelines” document must be followed; said document can be viewed here:
https://docs.google.com/document/d/1PKhHVvjYx7z2LNmfk-JkfosObsZRZTVFn8C81RYOVVA/edit?tab=t.0#heading=h.vrf4khtgogfz

Design Draft

1. The Hero Section (The "Qualified Filter")

This section must pass the "XY Test" within 5 seconds, explicitly naming the audience (Community Custodians), the outcome (bespoke intelligence hubs), and the mechanism (agentic framework).

- **H1 (The Value Prop):** The Agentic Intelligence Hub for Your Community.
- **H2 (The How):** Replace chaotic group chats and static newsletters with a programmable engine. Lume Centrix autonomously sources, summarizes, and vets content for your niche audience—delivered natively via Telegram and the Web.
- **Primary CTA:** Build Your Hub (Free Hobby Tier) — Low friction, instant gratification.
- **Secondary CTA:** Explore a Live Pulse — Interactive demo instead of a sales call.
- **Visual:** A split-screen mockup. Left side: The clean PWA interface. Right side: The slide-up Telegram Mini App (TMA) showing an expanded card. No people shaking hands.

2. Problem / Solution ("The Aha! Moment")

A brief section validating the pain of information overload, leading directly to your solution.

- **Heading:** Signal, not noise.
- **Copy:** Stop forcing your community to dig through unstructured social feeds or download clunky mobile apps. Lume Centrix provides a mission-led filter powered by AI and peer-review.
- **Visual:** A simple "Before & After" diagram.
    - *Before:* A messy web of YouTube, X, and RSS feeds pointing to a confused user.
    - *After:* The same feeds passing through the "Centrix Pipeline" into clean, categorized boards.

3. Core Capabilities (The "Pill" System)

Instead of long descriptions, we use scannable "pills" with icons to explain the technical moat.

- **Heading:** A full digital newsroom on autopilot.
- **Pills (Grid Layout):**

- 🤖 **7-Agent Pipeline:** Autonomous sourcing, video transcription (Whisper), and translation.
- 📱 **Zero App Development:** Your hub lives natively inside Telegram as an interactive Mini App. 100% push notification reach.
- 🧠 **Predictive Taste Graph:** The engine pre-ranks content sentiment before anyone even reads it, solving the "cold start" problem.
- 🛡️ **Publisher Vetting:** A built-in "Establishment Scorecard" tracks media objectivity and political leaning based on community ratings.
- 🗂️ **Multi-Board Architecture:** Configure up to 10 distinct feeds for News, Opportunities, Events, and more.

4. Pricing & Friction Reduction

To adhere to the "Orange Flag" rule, we skip the desperate "30-day money-back guarantee" and instead rely on a generous Hobby Tier and a "No Lock-In" clause.

- **Heading:** Start curating immediately.
- **Pricing Cards:**
  - **Hobby (Free):** Up to 2 Public Boards, Standard PWA Runtime, Basic Summarization.
  - **Elite Hub (Paid):** 10 Configurable Boards, Telegram Native Runtime (TMA), Full Agentic Suite (Transcribe/Translate), Advertiser Console.
- **Trust Micro-copy (Under the CTA):** Your data is yours. Export your community's Taste Graph and publisher ratings at any time. Cancel or downgrade to Hobby seamlessly.

5. Footer (Utility Layer)

Keep it disciplined and clean.

- Community Custodian Login
- Admin Console Documentation
- Terms of Service & Privacy
- Contact